

Jurnal Modul 13
Basis Data
13_NoSQL_Document_Database



Disusun Oleh
Nama : Radita Putri Nuraini
NIM : 103122400056

PROGRAM STUDI SOFTWARE ENGINEERING
FAKULTAS INFORMATIKA
TELKOM UNIVERSITY PURWOKERTO
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Disediakan dataset berikut:

```
db.toko.insertMany([
  {
    "name" : "Toko Berkah",
    "category": "Elektronik",
    "city" : "Jakarta",
    "address" : { "street": "Jl. Sudirman", "zipcode": "10220" },
    "products": [
      { "item": "TV", "price": 3500000, "stock": 10 },
      { "item": "Kulkas", "price": 2800000, "stock": 5 }
    ]
  },
  {
    "name" : "Toko Maju",
    "category": "Elektronik",
    "city" : "Bandung",
    "address" : { "street": "Jl. Asia Afrika", "zipcode": "40111" },
    "products": [
      { "item": "HP", "price": 1500000, "stock": 20 },
      { "item": "Laptop", "price": 8000000, "stock": 3 }
    ]
  },
  {
    "name" : "Toko Salim Bakery",
    "category": "Elektronik",
    "city" : "Jakarta",
    "address" : { "street": "Jl. Sudirman", "zipcode": "10220" },
    "products": [
      { "item": "Monitor LED", "price": 7700000, "stock": 11 },
      { "item": "Airfryer Panasonic", "price": 4850000, "stock": 7 }
    ]
  },
  {
    "name" : "Toko Sudinotech",
    "category": "Elektronik",
    "city" : "Surabaya",
    "address" : { "street": "Jl. Sudirman", "zipcode": "10220" },
    "products": [
      { "item": "RTX 55000", "price": 8500000, "stock": 20 },
      { "item": "Set PC Gaming Radeon", "price": 12000000, "stock": 9 }
    ]
  },
])
```

1. Tambahkan 1 toko baru bebas ke koleksi toko, dengan ketentuan:

- Minimal memiliki 2 produk di dalam array products
- Isi semua field: name, category, city, address, products

Hint:

```
db.toko.insertOne({
  "name" : "...",
  "category" : "...",
  "city" : "...",
  "address" : { "street" : "...", "zipcode" : "..." },
  "products" : [
    { "item" : "...", "price" : ..., "stock" : ... },
    { "item" : "...", "price" : ..., "stock" : ... }
  ]
})
```

Jawab :

```
> db.toko.insertOne({
  "name": "Toko Nusantara",
  "category": "Elektronik",
  "city": "Yogyakarta",
  "address": { "street": "Jl. Malioboro", "zipcode": "55213" },
  "products": [
    { "item": "Kipas Angin", "price": 350000, "stock": 15 },
    { "item": "Rice Cooker", "price": 450000, "stock": 8 }
  ]
})
```

```
< {
  acknowledged: true,
  insertedId: ObjectId('6a1931d69158e5207c293c6d')
}
```

2. Tampilkan semua toko yang berada di kota Jakarta.

Jawab :

```
> db.toko.find({ "city": "Jakarta" })
```

```
< {
  acknowledged: true,
  insertedId: ObjectId('6a1931d69158e5207c293c6d')
}
> db.toko.find({ "city": "Jakarta" })
< {
  _id: ObjectId('6a19302e9158e5207c293c69'),
  name: 'Toko Berkah',
  category: 'Elektronik',
  city: 'Jakarta',
  address: {
    street: 'Jl. Sudirman',
    zipcode: '10220'
  },
  products: [
    {
      item: 'TV',
      price: 3500000,
      stock: 10
    },
    {
      item: 'Kulkas',
      price: 2800000,
      stock: 5
    }
  ]
}
```

```

    }
  ]
}
{
  _id: ObjectId('6a19302e9158e5207c293c6b'),
  name: 'Toko Salim Bakery',
  category: 'Elektronik',
  city: 'Jakarta',
  address: {
    street: 'Jl. Sudirman',
    zipcode: '10220'
  },
  products: [
    {
      item: 'Monitor LED',
      price: 7700000,
      stock: 11
    },
    {
      item: 'Airfryer Panasonic',
      price: 4850000,
      stock: 7
    }
  ]
}
}

```

3. Tampilkan toko yang memiliki produk dengan price lebih dari 3.000.000.

Hint:

- Gunakan dot notation: products.price
- Gunakan operator: \$gt

Jawab :

```
> db.toko.find({ "products.price": { $gt: 3000000 } })
< {
  _id: ObjectId('6a19302e9158e5207c293c69'),
  name: 'Toko Berkah',
  category: 'Elektronik',
  city: 'Jakarta',
  address: {
    street: 'Jl. Sudirman',
    zipcode: '10220'
  },
  products: [
    {
      item: 'TV',
      price: 3500000,
      stock: 10
    },
    {
      item: 'Kulkas',
      price: 2800000,
      stock: 5
    }
  ]
}
```

```
  _id: ObjectId('6a19302e9158e5207c293c6a'),
  name: 'Toko Maju',
  category: 'Elektronik',
  city: 'Bandung',
  address: {
    street: 'Jl. Asia Afrika',
    zipcode: '40111'
  },
  products: [
    {
      item: 'HP',
      price: 1500000,
      stock: 20
    },
    {
      item: 'Laptop',
      price: 8000000,
      stock: 3
    }
  ]
}
{
  _id: ObjectId('6a19302e9158e5207c293c6b'),
  name: 'Toko Salim Bakery',
```

```
category: 'Elektronik',
city: 'Jakarta',
address: {
  street: 'Jl. Sudirman',
  zipcode: '10220'
},
products: [
  {
    item: 'Monitor LED',
    price: 7700000,
    stock: 11
  },
  {
    item: 'Airfryer Panasonic',
    price: 4850000,
    stock: 7
  }
]
}
{
  _id: ObjectId('6a19302e9158e5207c293c6c'),
  name: 'Toko Sudinotech',
  category: 'Elektronik',
  city: 'Surabaya',
```

```

address: {
  street: 'Jl. Sudirman',
  zipcode: '10220'
},
products: [
  {
    item: 'RTX 55000',
    price: 8500000,
    stock: 20
  },
  {
    item: 'Set PC Gaming Radeon',
    price: 12000000,
    stock: 9
  }
]
}

```

4. Ubah category Toko Berkah dari Fashion menjadi Aksesoris.
 Tambahkan 1 produk baru ke dalam array products milik Toko Maju menggunakan \$push.

Hint:

- Filter berdasarkan name: Toko Maju
- Gunakan operator \$push untuk menambah elemen ke array

Jawab :

```

> db.toko.updateOne(
  { "name": "Toko Berkah" },
  { $set: { "category": "Aksesoris" } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
test>

```

5. Hitung jumlah toko per city, urutkan dari kota yang paling banyak tokonya.
 Gunakan:
- \$group dengan \$sum: 1

- \$sort descending (-1)

Jawab :

```
> db.toko.aggregate([
  { $group: { _id: "$city", jumlah_toko: { $sum: 1 } } },
  { $sort: { jumlah_toko: -1 } }
])
< {
  _id: 'Jakarta',
  jumlah_toko: 2
}
{
  _id: 'Surabaya',
  jumlah_toko: 1
}
{
  _id: 'Bandung',
  jumlah_toko: 1
}
{
  _id: 'Yogyakarta',
  jumlah_toko: 1
}
test>
```

6. Hitung rata-rata harga produk per category.

Tahapan pipeline:

- \$unwind: '\$products' (bongkar array products dulu)
- \$group: kelompokkan per category, hitung \$avg dari products.price
- \$sort: urutkan sesuai kebutuhan

Jawab :

```
> db.toko.aggregate([
  { $unwind: "$products" },
  { $group: { _id: "$category", rata_rata_harga: { $avg: "$products.price" } } },
  { $sort: { rata_rata_harga: -1 } }
])
< {
  _id: 'Elektronik',
  rata_rata_harga: 5418750
}
{
  _id: 'Aksesoris',
  rata_rata_harga: 3150000
}
test>
```

7. Tambahkan 1 produk baru ke dalam array products milik Toko Maju menggunakan \$push.

Hint:

- Filter berdasarkan name: Toko Maju
- Gunakan operator \$push untuk menambah elemen ke array

Jawab :

```
> db.toko.updateOne(
  { "name": "Toko Maju" },
  { $push: { "products": { "item": "Tablet", "price": 3200000, "stock": 12 } } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
test> |
```

8. Tampilkan harga produk tertinggi (\$max) per city.

Hint:

- Unwind products terlebih dahulu
- Gunakan \$max pada products.price

Jawab :

```

> db.toko.aggregate([
  { $unwind: "$products" },
  { $group: { _id: "$city", harga_tertinggi: { $max: "$products.price" } } },
  { $sort: { harga_tertinggi: -1 } }
])
< {
  _id: 'Surabaya',
  harga_tertinggi: 12000000
}
{
  _id: 'Bandung',
  harga_tertinggi: 8000000
}
{
  _id: 'Jakarta',
  harga_tertinggi: 7700000
}
{
  _id: 'Yogyakarta',
  harga_tertinggi: 450000
}
test> |

```

9. Buat pipeline lengkap dengan 4 tahap:
- \$match — filter hanya toko category Elektronik
 - \$unwind — bongkar array products
 - \$group — hitung total stock per city
 - \$sort — urutkan dari total stock terbesar

Jawab :

```

> db.toko.aggregate([
  { $match: { "category": "Elektronik" } },
  { $unwind: "$products" },
  { $group: { _id: "$city", total_stock: { $sum: "$products.stock" } } },
  { $sort: { total_stock: -1 } }
])
< {
  _id: 'Bandung',
  total_stock: 35
}
{
  _id: 'Surabaya',
  total_stock: 29
}
{
  _id: 'Yogyakarta',
  total_stock: 23
}
{
  _id: 'Jakarta',
  total_stock: 18
}
test> |

```

10. Hapus toko yang memenuhi dua kondisi sekaligus:

- category adalah Fashion
- city adalah Surabaya

Hint:

- Gunakan \$and atau tulis kedua kondisi dalam satu objek filter

Jawab :

```

> db.toko.deleteMany({
  "category": "Fashion",
  "city": "Surabaya"
})
< {
  acknowledged: true,
  deletedCount: 0
}
test> |

```